How Autocracy Shapes the Social Sciences: Synthesized Evidence from an Agent Orchestra*

Pre-analysis draft — not for circulation

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Abstract

How does authoritarian rule shape the production of the social sciences and humanities? It is often assumed that we should observe large degrees of self-censorship and strategic content production among researchers in autocracies, in society-centered fields, yet there is little systematic evidence on this question. We look for signals of autocratic effects on social science across a wide range of different empirical implications, using a simulated research lab in the form of an agent orchestra. The orchestra uses a corpus of 2.7 million SSH articles from the Web of Science, matched to data on regime traits. Twenty-five teams of agents (designer, analyst, writer) each pre-register a distinct empirical implication of the self-censorship theory, organized into five sub-families: topic avoidance, framing neutrality, collaboration constraint, visibility suppression, and ideological alignment. In the final step we conduct a meta-analysis of the findings, discussing them in light of the overall theory. In this preliminary version we report results from 18 of the 25 teams that do not require external classifier APIs; the remaining seven LLM- and embedding-based teams are still in the queue. Across the 18 completed tests we find essentially no evidence of self-censorship: none of the primary coefficients on liberal democracy survives a within-family Bonferroni correction, and signs are mixed within and across sub-families. The agent-orchestra is offered as a scalable, reproducible and transparent way to evaluate theories with numerous different observable implications.

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Contents

1 Introduction

Autocratic regimes are increasingly making an imprint on the global production of science. China now produces more scientific articles than the United States (citation), and authoritarian countries like Saudi Arabia, Singapore and Qatar are investing heavily in academic research. How does science under autocracy differ from in a democracy? This paper considers this question in the context of the society-centered sciences, the social sciences and humanities (SSH fields). We investigate empirical implications of the widespread notion that we should observe self-censorship and strategic scientific content production in autocracies.

The notion that researchers in SSH fields in autocracies should alter their behavior, regarding what they study, how they frame their findings, who they collaborate with and how visible their work becomes has a long list of different implications – statistical signals that we should observe across a published academic work. The theory at hand thus yields a large number of different empirical implications, and it would be less informative to just focus on one. We therefore use a novel research approach that allows us to investigate several different implications under one coherent research framework: An agent-orchestra based meta-analysis. This starts from a pre-processed dataset of roughly 2,7 million SSH articles from the Web of Science (a random sample constituting around 70% of the total corpus), in the period 1970-2020 for 167 countries. We then set up 25 "research teams"', consisting of a study designer, an analyst agent, and a report writer. Each team comes up with a specific hypothesis derived from the over-arching theory that scientists react strategically to political context when doing SSH work in autocratic regimes (more so than in democracies). Each hypothesis (mechanism) is then reviewed by the PI (the lead author) and organized in theoretical sub-families, and hypotheses are pre-registered and time-stamped.[1] Each team then proposes a specific design to test the respective hypotheses, and these are also reviewed and improved where needed by the PI.

The over-arching theory is driven by the core assumption that researchers will strategically alter their research behavior to minimize professional consequences relating to the regime-context they operate under. From this general assumption, we explicate five distinct families of theoretical mechanisms that can in themselves yield empirical hypotheses to be tested. These are: Topic avoidance (researchers avoid certain sensitive topics), framing neutrality (researhcers frame their work in less politically sensitive ways), collaboration constraints (researchers avoid international networks that may attract negative regime-attention), visibility suppression (that self-censored output should generate less visible

[1] All 25 hypotheses and analysis plans were locked before any analyst session was run. Each team's preregistration document — including the verbal hypothesis, predicted sign, primary specification, robustness checks, and Bonferroni family — is stored at `teams/team_NN/preregistration.md` in the project repository at `https://github.com/torewig/autocracy-agent-orchestra` and is time-stamped by Git commit `960525f` ("Pre-registration: 25 teams," 6 May 2026). No outcome-driven changes have been made to the primary specifications since that commit.

work), and ideological alignment (active production of regime consonant and avoidance of regime-dissonant framings). The team then derived specific testable hypotheses for each of these families.

2 Related literature

The impact of political regimes on academia and science more generally is a topic that has spurred much discussion across fields. In philosophy of science, it is often taken for granted that there is a link between "open societies"' and thriving science (**??**). In political science, several contributions highlight how universities and academia can be in tension with autocratic regimes (**????**), and several single-country studies focus specifically on how autocrats crack down on academia (**???**). A quite recent literature discusses how the new wave of authoitarian populism, signified by autocrats such as Trump, and Orban (Hungary) antagonize and target left-leaning academic fields such as gender-studies (**?**).

Political regimes (democracy-autocracy) is often studied as a determinant of scientific output, both in general (**?**) and for the social sciences and humanities more specifically (**??**). Studies indicate that the social sciences and humanities suffer pernicious effects under autocracy (**??**). Cross-country work indicates that the democratization leads to a substantial increase in the dissemination and impact of social science research (**?**). Other studies also document positive correlations between democracy scores and greater scholarly output and citations **?**.

The drop in activity and impacts in the social sciences likely reflects growing pressures on academia by autocrats. Autocrats dispose of a range of tools that can be used to crack down on and align the social sciences with their (**??**). Autocrats can pull funding for specific fields and targeted institutional measures that ensure control. Notable examples include Russia's 2022 announcement of the discontinuation of sociology and political science in some universities and Hungary's expulsion of the Central European University, known for its social science focus (**??**). In autocratic regime contexts, researchers must navigate blurred "red lines" to avoid topics deemed politically sensitive, such as authoritarianism or public protests. Other autocratic regimes use softer tools such as state-controlled funding to incentivize research that aligns with government ideology (**??**). In China, for instance, major funding sources for the social sciences and humanities are controlled by branches of the Communist Party (**?**).

In spite of this growing literature considering links between autocratic regimes and the social sciences and what and how researchers in these regimes study, the contents of social science research. If researchers actively self-censor in autocracies (compared to democracies) we should expect systematic differences in the contents and ideological lean of social science and humanities research. This is the gap we seek to fill in this study.

By considering how autocracy affects the contents, framings and direction of the social sciences and humanities, under the driving assumption that researchers self-censor, we add to several literatures. First, we contribute to the growing political science literature on the relaitonships between autocracy-democracy and higher education and science. This is now expanding from focusing on universities as sources of dissent and liberal values (**??**) to how autocracy affects the intellectual products that eminate from universities. Second, we advance the ongoing discussion in the philosophy of science and science studies on how democracy matters for science (**?**), by showing concrete empirical results. Third, we add to the bibliometric literature that has studied the impact of regime type on scientific output volume and citations. We go beyond these measures to capture effects on deeper dimensions of scientific contents.

3 Theory: Self-Censorship under Autocracy

We start from a set of assumptions about the motivations of researchers in the social sciences and humanities. We assume that these are driven by career incentives (to gain prominence and tenure, and get research funding) and scientific motivations (to move the literature forward and address puzzles). Furthermore, we assume that they also have to take into account the constrints placed upon them by the autocratic regime and its institutons, and act strategically within this context. While researchers want to advance their career and find things out, they also value their physical, economic and social security, and would – ceteris paribus – seek to avoid different types of punishment they can face from the regime. They respond rationally to both negative and positive incentive structures influenced by the regime.

On the regime side, we assume that the ruler and his support coalition pursues political survival and want to supress threats from the opposition. To attain this, they want to minimize research dissemination and teaching that goes against official narratives and spreads potentially regime-threatening ideas relating to liberalism, democracy, human rights, minority protections and so fort, and they also want to minimize regime-criticism both direct and indirect.

The core theoretical claim is that researchers in autocracies may (relative to those in democracies) face career costs for politically sensitive topics, methods, collaborations, and framings.

This broad assumption is of the kind that yields a long list of empirical implications rather than one core test. Since there allways will be some uncertainty both on the regime side and the researcher side relating to what type of content will generate a regime reaction, we expect the signals of self-censorship behavior to be wide-ranging, occurring in multiple different domains, and often context dependent. This particular theoretical characteristic argues for testing several different implications, on many different content-outcomes.

We broadly outline five families of mechanisms that may result from the core theoretical assumption:

Topic avoidance: One mechanism-category has to do with researches de-prioritizing topics that (under some uncertainty) may come with regime-imposed costs. These may be topics that have to do with democratic values or democracy specifically, poor govenrance or corruption or political philosophies that engender criticism of power. The strategic behavior may occur at a very “macro"' level, where researchers select out of entire disciplines where these topics may arise (like political science), or be more fine-grained e.g. when researchers avoid working on “democracy"' per se. We register 7 hypotheses in this category. Examples include: That democracies should have a higher share of political-content keywords in published articles, that democracies should produce more work in politically sensitive fields (e.g. PoliSci), that they should more often mention democracy, human rights etc., and that they should have a higher share critically examining own-country institutions.

Framing neutrality: A second category relates to how researchers frame their findings. When there is uncertainty relating to regime-reactions, researchers may use less clear-cut language, hedge their bets, depoliticize conclusions and adopt more technocratic language in their conclusions and problem descriptions. We propose several hypotheses relating to such framing neutrality. We registered 6 hypotheses in this category, including: That autocracies should have a higher rate of “hedging"' words in bastracts, that they should have less normative language, and a lower share ending with normative conclusions.

Collaboration constraint: International research networks may threaten autocratic regimes and disseminate regime-critical ideas and frameworks through research networks. If this is seen and reacted to by autocratic regimes, one would expect researchers to avoid strong international visibility, deep engagement with international networks, and particularly with partners from democratic regimes. We develop several different hypotheses that would trace this self-selection out of international networks. For example we register the hypothesis that researchers in autocracies collaborate less with researchers in democracies, have less institutional diversity in collaboration and have smaller research teams.

Visibility suppression: A third category relates to lowered visibility of work that is (assumed) self-censored. This arguments trades on the assumption that self-censorship acts as a perversion in the machinery of scientific output, in such a way that the work that is constrained by self-censorship is of lower quality than more uncensored work. Examples of hypotheses in this category are that autocracies should have normalized citation impact, that the citation penalty for autocracies should interact with sensitive topics, and that autocracies should have higher citation variance.

Ideological alignment: In the final category of mechanisms, we present hypotheses under the assumption that researchers avoid semantic content that is dissonant with regime ideology. For example that autocracies should have a higher share of work that is higher share of work that has a neutral framing of political topics when compared to work that has a liberal political framing, that autocracies should have a higher share of work implicitly legitimating their own political system, and that they should have a higher share of work that has an anti-liberal framing.

3.1 Hypotheses and significance thresholds

Table 1 lists the five theory sub-families, the teams assigned to each, and the Bonferroni-corrected significance threshold for each family.

Table 1: Theory families, team assignments, and Bonferroni thresholds

Sub-family	Teams	k	Adjusted α ($0.05/k$)
topic-avoidance	01–07	7	$p < 0.0071$
framing-neutrality	08–13	6	$p < 0.0083$
collaboration-constraint	14–18	5	$p < 0.0100$
visibility-suppression	19–22	4	$p < 0.0125$
ideological-alignment	23–25	3	$p < 0.0167$
Total	01–25	**25**	

All hypotheses are listed in table **??** in section **??**below.

4 Methodological approach: Synthetized evidence from an AI-agent orchestra

The approach taken in this paper is unconventional. Rather than proposing a detailed research design testing one specific implication of the theory, and working as hard as possible to shore up the causal assumptions of that specific design, we take an approach that is more capable of accomodating information from several different tests across a wide range of outcome variables. More specifically, we set up an AI agent orhcestra (using ClaudeCode, and Anthropic's Opus and Sonnet models) that collaborates with the Principal Investigator (lead author) in several steps. The orchestra consists of 25 "research teams"' consisting of i) a study designer that develops a clear hypothesis, and proposes an analysis plan, ii) an data analyst that executes the pre-registered analysis in R, a iv) Writer that drafts up a short research report of the findings, a iii) reviewer that provides feedback on the report, and a synthesizer that proposes a model for cross-team theory evaluation. In each step, the PI is providing feedback, redirecting the team, reviewing code and results, and proposes new directions. This is a classic case of "agent-orchestration"' which is developing as a new methodology in social science data analysis (**?**).

The approach is described in greater detail below. But it requires a more in-depth justification here. Why use this approach when more standard approaches are available? First, this is a highly useful method for the specific purpose at hand, which is to test a theory that has an almost inumerable number of concrete empirical implications, since the core theoretical mechanism (researcher self-censorship and strategic behavior) is unobservable, and the many behaviors that are consistent with it and thus the potential universe of statistically observable signals in published work, almost infinite. Second, it is a low-cost way to push the research frontier on a "research project"' scale with relatively low investment. Investigating a large number of hypotheses with heavy-duty data analysis would require a medium-sized research team of several postdocs and PhDs working over a long period. Using an agent-orchestra speeds and scales up this process, providing plausibly similar results. Third, the chosen approach here includes a heavy emphasis on "human in the loop"' elements, where the lead author (as PI) reviews all the code, provides feedback on all analysis plans and quality checks all output, in the same way as a lead researcher would do on a research project leading a team of junior scholars.

The approach undoubtedly comes with downsides. First, it means that some of the input to the project (e.g. design proposals, coding decisions, operationalizations, etc. comes from AI), which can hallucinate and make unsupervised mistakes. To ameliorate these issues, the PI has read and tested the code, and supervised all stages of the process. All elements of the analysis is verified by the PI and can be replicated from the raw data. To ensure full transparency, all code, all prompts and agent-steering documents, and all analysis logs and pre-analysis plans are available on github.

5 Data and Corpus

All teams work from the same data source, which is a bibliometric corpus of articles published in the Web of Science (downloaded from Web of Science through their API in 2021, by the PI and co-authors). This corpus is then filtered on social science and humanities articles (by the field-categories listed in the web of science and noted on each article). The final corpus includes 2,709,224 SSH articles, in the years 1970-2023. Each article is then broken down into article-rows that can be assigned to countries. When articles are co-authored, we count each co-authoring country as 1 article for that country. All articles contain information on title, publication time, abstract, journal, subject keywords, field classification, institutions of the authors and addresses. These addresses are matched to countries with a very high success rate.

These article-data are then matched to information on political regimes, using the Varieties of Democracy dataset (?). As the main predictor variable we use the Liberal Democracy index, which should most closely line up with the theory that relates to academic self-censorship based on threats to free speech and professional sanctions. The liberal democracy index emphasizes civil rights and free speec (and academic freedom) in addition to the core electoral components of democracy and is thus well suited as our main independent vairable. It ranges between 0 and 1. As an alternative regime indicator we use the Lexical Index of Democray (**?**) in robustness tests. We also provide a set of standard controls, that each team will use, log(Gdp per capita) and log(population)

To make sure that we are not just capturing differences between those countries that publish nothing in a given field and those that publish more than 0, we preprocess the data to only include countries that publish at least 10 articles in the SSH corpus in a given year. The article corpus data, with the above listed variables, is fully locked prior to any analysis and can not be manipulated by the agents.

6 Agent-Orchestra Methodology

6.1 Pipeline overview

The multi-agent pipeline consists of five sequential agent types. Each team is assigned one instance of each agent. All designs are pre-registered before any analysis is run.

Figure 1 illustrates the full pipeline.

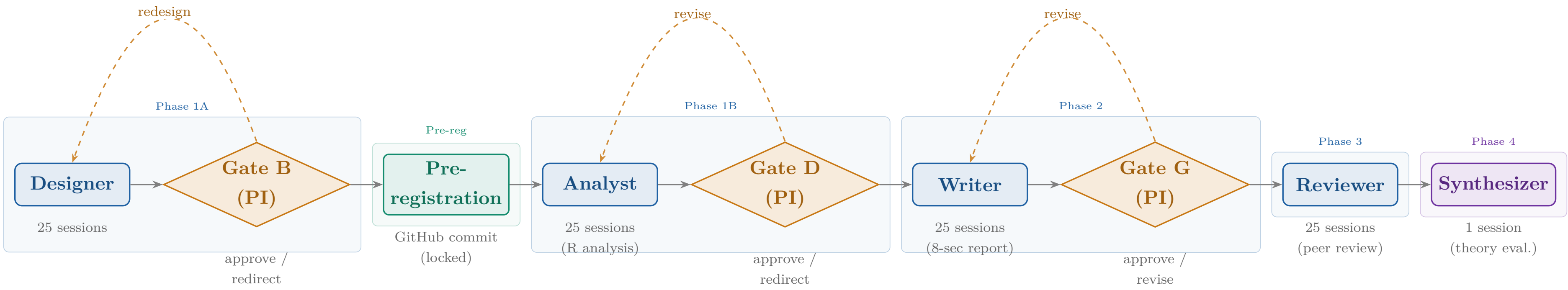

Figure 1: Agent-orchestra pipeline. **Blue**: agent sessions (Claude Sonnet/Haiku); **amber**: PI gate reviews; **teal**: pre-registration milestone; **violet**: synthesis. Dashed arrows indicate redesign/revision loops possible at each gate. Current status: pre-registration complete (gate B passed); analyst sessions pending.

6.2 Agent roles

Each team runs four agents in sequence. The **Designer** develops the research question, operationalization, and analysis plan within a pre-assigned theoretical domain, producing `rq.md` and `analysis_plan.md`. The **Analyst** executes the pre-registered analysis in R against the locked corpus and writes `analysis.R`, the figures, and a machine-readable `primary_results.json`. The **Writer** drafts the team report following a standardized 8-section template (`report.md`), and the **Reviewer** — a separate agent instance — peer-reviews that report against rubric criteria (`peer_review.md`).

Once all 25 teams have completed this pipeline, a single **Synthesizer** agent reads every team report and peer review and produces a cross-team theory evaluation (`synthesis/theory_evaluati`
that aggregates the sub-family verdicts against the overarching self-censorship theory.

6.3 Pre-registration

All 25 teams' hypotheses and analysis plans were committed to GitHub (commit `960525f`) on 6 May 2026, before any analyst session was run. The timestamp and Git commit hash constitute the pre-registration record. Bonferroni adjustment is computed programmatically post-analysis using `scripts/bonferroni_adjust.R`.

6.4 Estimation strategy

Primary specification (all teams, country-year level). Two-way fixed effects (TWFE):

$$Y_{it} = \beta \cdot \texttt{v2x\_libdem}_{it} + \gamma_1 \log(\texttt{gdppc}_{it}) + \gamma_2 \log(\texttt{pop}_{it}) + \alpha_i + \delta_t + \varepsilon_{it} \tag{1}$$

where α_i = country FE, δ_t = year FE, SE clustered by `iso3`.

Secondary specification (pooled OLS). Year FE only (no country FE), SE clustered by `iso3`. Checks whether the within-country estimate is consistent with the cross-sectional pattern.

Robustness checks. All teams pre-specify RC1 (sample restriction or alternative operationalization) and RC2 (replace `v2x_libdem` with `lied_binary`).

7 Project Disposition and Current Stage

7.1 Phase 0 — Data preparation (Complete)

- Raw WOS SSH records downloaded and parsed: 3,189,557 rows
- De-duplicated to 2,709,224 unique articles (`wos_id`)

- V-DEM political data merged at `iso3` × `year` level
- Country coverage: 99.98% match rate
- Controls added: log GDP per capita, log population
- Output: `data/agent_corpus.rds` (locked; not tracked in Git)

7.2 Phase 1A — Designer sessions (Complete)

- 25 Designer agent sessions run (Claude Sonnet)
- Each session: read brief + data description, develop RQ within domain
- Output: `rq.md` + `analysis_plan.md` per team
- Uniqueness check enforced: no two teams share outcome + unit + strategy

7.3 Gate B — PI review (Complete, 6 May 2026)

- All 25 teams reviewed by PI
- Gate criteria: clear H1, specified controls, ≥ 2 robustness checks, uniqueness verified, self-censorship framing coherent
- Revisions: Teams 15–25 revised post-review; Team 21 fully redesigned (original interaction design rejected; replaced with `prop_ever_cited`)
- All 25 teams approved

7.4 Pre-registration (Complete, 6 May 2026)

- Script: `scripts/preregister.ps1`
- Creates timestamped `preregistration.md` = RQ + analysis plan
- Git commit: `960525f` ("Pre-registration: 25 teams")
- Repository: GitHub (remote `origin/main`)
- All 25 hypotheses and analysis plans locked

7.5 Phase 1B — Analyst sessions (*Pending*)

- 25 Analyst agent sessions to be run
- Each session: read `preregistration.md`, execute analysis in R, write `primary_results.json` and produce figures
- Simple teams (no API): estimated ~2h per session
- Complex teams (LLM API): estimated ~4–8h per session, $2–500 API budget per team
- Planned order: Simple and Medium teams first (01–04, 07–09, 11, 13–17, 19, 22); Complex teams in parallel (05, 06, 10, 12, 23–25)

7.6 Gate D — PI review of results (*Pending*)

- Review all 25 figures and `primary_results.json` files
- Criteria: analysis executed as pre-registered, results interpretable
- Option to redirect teams with implementation errors only (no outcome-driven changes)

7.7 Phase 2 — Writer sessions (*Pending*)

- 25 Writer agent sessions (8-section standardized report template)
- Bonferroni adjustment computed first: `scripts/bonferroni_adjust.R`

7.8 Gate G — PI review of reports (*Pending*)

7.9 Phase 3 — Reviewer sessions (*Pending*)

- Peer review of each report by a separate Reviewer agent instance

7.10 Phase 4 — Synthesis (*Pending*)

- Synthesizer agent reads all 25 reports + peer reviews
- Output: `synthesis/theory_evaluation.md`
- One verdict per sub-family: Supports / Partially supports / Mixed / Does not support

8 Results

8.1 Coverage so far

Of the 25 pre-registered teams, **18 have completed analyst sessions** at the time of writing: all teams in the topic-avoidance, framing-neutrality, collaboration-constraint, and visibility-suppression sub-families that rely only on the bibliometric corpus and V-Dem. The remaining **seven teams** — 05, 10, 12 (LLM classifiers of framing), 06 (text embeddings), and 23–25 (the entire ideological-alignment sub-family, LLM classifiers of regime-consonant content) — require external API calls and are queued pending budget sign-off. Section 8 therefore reports a preliminary snapshot covering 18 of 25 pre-registered implications, organized by the four sub-families for which results are currently in.

8.2 Main coefficients by sub-family

Figure 2 summarizes the primary estimate from each of the 18 completed teams. Each row is one team's pre-registered test: the point is the standardized estimate $\hat{\beta}/\widehat{\mathrm{SE}}$ of the coefficient on `v2x_libdem` from the TWFE specification (country and year fixed effects, log GDP per capita and log population as controls, standard errors clustered by country), and the horizontal bar is the 95% confidence interval in the same standardized scale. The dashed vertical lines mark the nominal ± 1.96 threshold; the dotted lines mark the family-specific Bonferroni-corrected 5% threshold ($\alpha = 0.05/k$ within each sub-family).

The pattern across the 18 tests is clear: **no team's primary coefficient survives its family-specific Bonferroni threshold**, and only one team (Team 22, citation Gini) clears the nominal $p < 0.05$ line. Even that single nominal exception goes in the *opposite* direction to its hypothesis. Within sub-families, signs are mixed for topic avoidance and framing neutrality; the four collaboration-constraint estimates (Teams 14, 15, 17, 18) all point in the theory-predicted direction but none of them reaches significance; the visibility-suppression estimates match the predicted sign but cluster tightly around zero.

8.3 Consistency across specifications

[TODO: TWFE vs. pooled OLS comparison once all 25 teams' pooled-OLS secondary results are tabulated.]

8.4 Robustness

[TODO: RC1 and RC2 results summary.]

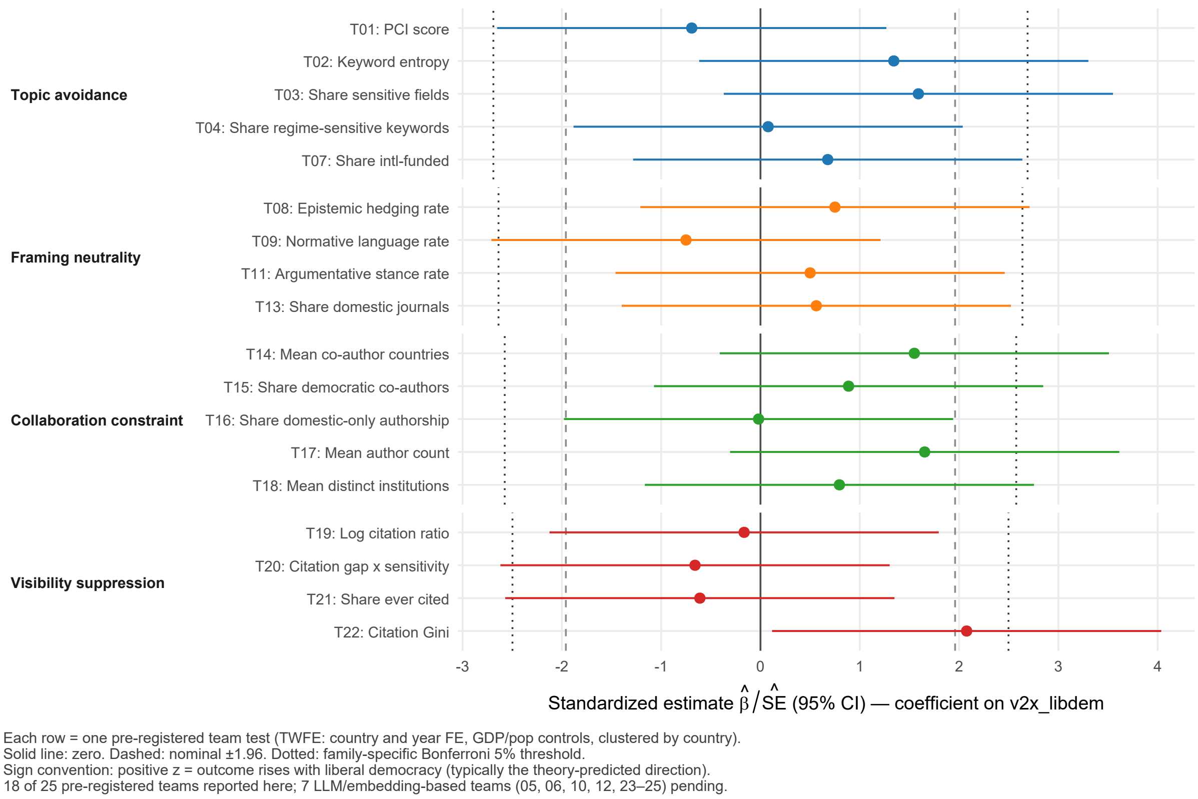

Figure 2: Main coefficients on `v2x_libdem` for the 18 completed teams, organized by theory sub-family. Standardized as $\hat{\beta}/\widehat{\mathrm{SE}}$ with 95% confidence intervals. Solid line: zero. Dashed: nominal ± 1.96. Dotted: family-specific Bonferroni 5% threshold. Positive values indicate that the outcome rises with liberal democracy, which corresponds to the theory-predicted direction for most — but not all — teams (see each team's `preregistration.md` for the expected sign).

9 Discussion

[To be written after synthesis.]

Key points (skeleton).

- Which sub-families show most consistent evidence?
- Heterogeneity: where is self-censorship concentrated?
- Limitations: observational, cannot rule out selection, corpus skewed to WOS-indexed output
- Implications for academic freedom research

10 Conclusion

[To be written after analysis.]

A Team roster and file map

Table 2: Team file structure (one row per team)

Team	Sub-family	`rq.md`	`analysis_plan.md`	`preregistration.md`
01–07	topic-avoidance	✓	✓	✓
08–13	framing-neutrality	✓	✓	✓
14–18	collaboration-constraint	✓	✓	✓
19–22	visibility-suppression	✓	✓	✓
23–25	ideological-alignment	✓	✓	✓

B Full list of pre-registered hypotheses

Table 3 reproduces the verbatim H1 sentence locked in each team's `preregistration.md` at Git commit `960525f` (6 May 2026). The condensed version in Table **??** (main text, §**??**) shortens the same sentences for landscape display; this appendix preserves the exact text as pre-registered.

Table 3: Full pre-registered hypotheses, all 25 teams

T	Sub-family	Outcome variable	Hypothesis (verbatim from preregistration.md)
01	topic-avoidance	`pci_mean`	Countries with lower Liberal democracy levels will exhibit lower mean PCI scores in SSH abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
02	topic-avoidance	`keyword_entropy_cy`	Countries with lower Liberal democracy levels will exhibit lower Shannon entropy of author-supplied keyword distributions at the country-year level, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
03	topic-avoidance	`share_sensitive_fields`	Countries with lower Liberal democracy levels will produce a lower share of SSH output in politically sensitive disciplines, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
04	topic-avoidance	`share_sensitive`	Countries with lower Liberal democracy levels will have a lower share of SSH articles containing regime-sensitive terms in titles, author keywords, and abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
05	topic-avoidance	`share_critical_framing`	Countries with lower Liberal democracy levels will exhibit a lower share of SSH articles critically examining their own governance and institutions, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.

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T	Sub-family	Outcome variable	Hypothesis (verbatim from preregistration.md)
06	topic-avoidance	`mean_semantic_diversity`	Countries with lower Liberal democracy levels will exhibit lower mean pairwise semantic diversity of SSH abstracts within country-field-year clusters, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
07	topic-avoidance	`share_intl_funded`	Countries with lower Liberal democracy levels will have a lower share of SSH articles acknowledging international funding agencies, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
08	framing-neutrality	`hedging_rate`	Countries with lower Liberal democracy levels will exhibit higher mean epistemic hedging rates in SSH abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
09	framing-neutrality	`normative_rate`	Countries with lower Liberal democracy levels will exhibit lower mean normative language rates in SSH abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
10	framing-neutrality	`share_technocratic`	Countries with lower Liberal democracy levels will exhibit a higher share of SSH abstracts classified as purely technocratic (no political references), after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
11	framing-neutrality	`stance_rate`	Countries with lower Liberal democracy levels will exhibit lower mean rates of first-person argumentative stance phrases in SSH abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.

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T	Sub-family	Outcome variable	Hypothesis (verbatim from preregistration.md)
12	framing-neutrality	`share_normative_conclusion`	Countries with lower Liberal democracy levels will exhibit a lower share of SSH abstracts ending with an explicit normative or policy conclusion, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
13	framing-neutrality	`share_domestic_journal`	Countries with lower Liberal democracy levels will publish a higher share of SSH output in domestically dominated journals, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
14	collaboration-constraint	`mean_n_coauthor_countries`	Countries with lower Liberal democracy levels will exhibit a lower mean number of distinct co-author countries per SSH article, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
15	collaboration-constraint	`share_libdem_coauth`	Countries with lower Liberal democracy levels will have a lower share of SSH articles co-authored with researchers from liberal democracies (v2x_libdem > 0.5), after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
16	collaboration-constraint	`share_domestic_only`	Countries with lower Liberal democracy levels will exhibit a higher share of SSH articles with exclusively domestic authorship, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
17	collaboration-constraint	`mean_n_authors`	Countries with lower Liberal democracy levels will exhibit a lower mean number of authors per SSH article, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.

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T	Sub-family	Outcome variable	Hypothesis (verbatim from preregistration.md)
18	collaboration-constraint	`mean_n_institutions`	Countries with lower Liberal democracy levels will exhibit a lower mean number of distinct author institutions per SSH article, after controlling for country fixed effects, year fixed effects, log GDP per capita, log population, and mean co-author country count.
19	visibility-suppression	`log_cite_ratio_mean`	Countries with lower Liberal democracy levels will exhibit a lower mean normalized citation ratio for SSH articles, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
20	visibility-suppression	`log1p(cite_ratio)`	The citation penalty associated with lower Liberal democracy levels will be larger for politically sensitive articles than for non-sensitive articles, as indicated by a positive coefficient on the v2x_libdem × sensitive_flag interaction term, after controlling for country fixed effects, year fixed effects, field fixed effects, log GDP per capita, and log population.
21	visibility-suppression	`prop_ever_cited`	Countries with lower Liberal democracy levels will exhibit a lower share of SSH articles that receive at least one citation within a country-year, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
22	visibility-suppression	`gini_tot_cites`	Countries with lower Liberal democracy levels will exhibit a higher Gini coefficient of citations across SSH articles, after controlling for country fixed effects, year fixed effects, log GDP per capita, log population, and log article volume.

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T	Sub-family	Outcome variable	Hypothesis (verbatim from preregistration.md)
23	ideological-alignment	`share_apolitical`	Countries with lower Liberal democracy levels will exhibit a lower share of SSH abstracts with liberal/neutral political-economy framing (NEUTRAL + NONE) and a higher share with ideologically aligned framing (LEFT or RIGHT-CONSERVATIVE), after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
24	ideological-alignment	`share_legitimating`	Countries with lower Liberal democracy levels will exhibit a higher share of SSH abstracts that actively legitimize or endorse the current political system, state authority, or leadership, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
25	ideological-alignment	`share_anti_liberal`	Countries with lower Liberal democracy levels will exhibit a higher share of SSH abstracts that actively frame liberal democracy, Western political institutions, or international human rights norms as fundamentally flawed or illegitimate, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.

C Per-team mini research plans

Each block below summarises the pre-registered design for one team, as locked at Git commit `960525f`. Hypotheses, controls and robustness labels are extracted programmatically from `teams/team_NN/preregistration.md` by `scripts/make_appendix_team_material.R`; the source of truth for any disagreement is the preregistration file itself, not this table.

Team 01 — *topic-avoidance*

Hypothesis	Countries with lower Liberal democracy levels will exhibit lower mean PCI scores in SSH abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`pci_mean` — country-year mean PCI score (share of abstract text matching political-content keyword dictionary)
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	Alternative regime measure (binary); Alternative regime measure (ordinal); Weighted aggregation; Narrow dictionary

Team 02 — *topic-avoidance*

Hypothesis	Countries with lower Liberal democracy levels will exhibit lower Shannon entropy of author-supplied keyword distributions at the country-year level, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`keyword_entropy_cy` — Shannon entropy of the country-year author-keyword distribution
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	Binary regime measure; Unweighted model; Full sample (1970-2023); Type-token ratio alternative; Subject-field restricted sample

Team 03 — *topic-avoidance*

Hypothesis	Countries with lower Liberal democracy levels will produce a lower share of SSH output in politically sensitive disciplines, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_sensitive_fields` — country-year share of output published in politically sensitive WOS disciplines
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1 — Expanded field list; RC2 — History-included; Alternative regime measure; Fractional logit; Lagged IV; Restricted time window

Team 04 — *topic-avoidance*

Hypothesis	Countries with lower Liberal democracy levels will have a lower share of SSH articles containing regime-sensitive terms in titles, author keywords, and abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_sensitive` — country-year share of articles whose abstracts mention regime-sensitive terms (democracy, rights, corruption, protest)
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE
Robustness checks	Pre-specified robustness checks (RC1–RC2) required by Gate B; RC1 — Title + keywords only; RC2 — Alternative regime measure; Additional robustness checks; Regime type categories; Political science / sociology subsample

Team 05 — *topic-avoidance*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a lower share of SSH articles critically examining their own governance and institutions, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_critical_framing` — country-year share of abstracts critically examining own-country governance (LLM-classified)
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE
Robustness checks	Alternative regime measure; Regime-type heterogeneity; Field restriction; English-only abstracts; Fractional logit; Alternative threshold

Team 06 — *topic-avoidance*

Hypothesis	Countries with lower Liberal democracy levels will exhibit lower mean pairwise semantic diversity of SSH abstracts within country-field-year clusters, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`mean_semantic_diversity` — country-year mean pairwise cosine distance of abstract embeddings within field-year clusters
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1 - Country-field-year level

Team 07 — *topic-avoidance*

Hypothesis	Countries with lower Liberal democracy levels will have a lower share of SSH articles acknowledging international funding agencies, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_intl_funded` — country-year share of articles acknowledging international funding agencies
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1 — Alternative regime measure; RC2 — Restrict to grant_agencies field only

Team 08 — *framing-neutrality*

Hypothesis	Countries with lower Liberal democracy levels will exhibit higher mean epistemic hedging rates in SSH abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`hedging_rate` — country-year mean rate of epistemic hedging phrases per abstract word
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	Binary democracy indicator; Discipline restriction; Extended hedging dictionary; Post-1990 subsample; RC1 - Restrict to English-language abstracts; RC2 - Narrow hedging dictionary

Team 09 — *framing-neutrality*

Hypothesis	Countries with lower Liberal democracy levels will exhibit lower mean normative language rates in SSH abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`normative_rate` — country-year mean rate of normative/prescriptive terms per abstract word
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1 - Restricted dictionary; R1 - Replace v2x_libdem with lied_binary; R2 - Restrict dictionary to terms with minimal PCI overlap...; R3 - Restrict corpus to politically sensitive disciplines...; R4 - One-year within-country lag of v2x_libdem; R5 - Replace mean_norm_rate with median_norm_rate

Team 10 — *framing-neutrality*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a higher share of SSH abstracts classified as purely technocratic (no political references), after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_technocratic` — country-year share of abstracts classified as exclusively technocratic (LLM)
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1 — Alternative regime measure; RC2 — Restrict to politically sensitive fields; RC3 — English-only abstracts subsample; RC4 — Alternative classification threshold

Team 11 — *framing-neutrality*

Hypothesis	Countries with lower Liberal democracy levels will exhibit lower mean rates of first-person argumentative stance phrases in SSH abstracts, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`stance_rate` — country-year rate of 1st-person argumentative stance phrases
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1 — Restrict to English-language abstracts (critical); RC2 — Alternative regime measure: lied_binary; R3 — Disciplinary subsample: political science and sociology; R4 — Expanded phrase list

Team 12 — *framing-neutrality*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a lower share of SSH abstracts ending with an explicit normative or policy conclusion, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_normative_conclusion` — country-year share of abstracts ending with an explicit normative/policy conclusion (LLM)
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1 — Alternative regime measure; RC2 — Restrict to politically sensitive fields

Team 13 — *framing-neutrality*

Hypothesis	Countries with lower Liberal democracy levels will publish a higher share of SSH output in domestically dominated journals, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_domestic_journal` — country-year share of articles published in domestically dominated journals
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1 - Vary domesticity threshold; RC2 - Restrict to small and medium SSH producers; RC3 - Alternative regime measure; R4 - Restrict to 1990–2005 (limit confounding from...

Team 14 — *collaboration-constraint*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a lower mean number of distinct co-author countries per SSH article, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`mean_n_coauthor_countries` — country-year mean number of distinct co-author countries per article
Predictor	`v2x_libdem`
Controls	baseline (log GDP per capita, log population, country FE, year FE)
Robustness checks	RC1: Restrict to multi-author articles; RC2: Alternative regime measure; R1: Binary regime IV; R2: Median instead of mean; R3: Log-transformed outcome; R4: Intensive margin — internationally co-authored articles only

Team 15 — *collaboration-constraint*

Hypothesis	Countries with lower Liberal democracy levels will have a lower share of SSH articles co-authored with researchers from liberal democracies (v2x_libdem > 0.5), after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_libdem_coauth` — country-year share of articles with at least one co-author from a liberal democracy
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE
Robustness checks	RC1 — Vary the liberal democracy threshold; RC2 — Alternative regime measure; Alternative IV — lied_binary; Alternative co-author threshold — v2x_libdem > 0.7; Alternative IV + co-author threshold; Opportunity-set normalisation

Team 16 — *collaboration-constraint*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a higher share of SSH articles with exclusively domestic authorship, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_domestic_only` — country-year share of articles with exclusively domestic authorship
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE
Robustness checks	RC1 — Restrict to multi-author articles; RC2 — Alternative regime measure

Team 17 — *collaboration-constraint*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a lower mean number of authors per SSH article, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`mean_n_authors` — country-year mean author count per article
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE
Robustness checks	RC1: Field fixed effects; RC2: Alternative regime measure; R1: Alternative IV; R2: Log outcome; R3: Control for international co-authorship rate; R4: Restrict to post-1990

Team 18 — *collaboration-constraint*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a lower mean number of distinct author institutions per SSH article, after controlling for country fixed effects, year fixed effects, log GDP per capita, log population, and mean co-author country count.
Outcome variable	`mean_n_institutions` — country-year mean number of distinct author institutions per article
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`, `mean_n_coauthor_countries`; plus country FE + year FE
Robustness checks	RC1 — Restrict to multi-author articles; RC2 — Alternative regime measure; Log-transformed outcome; Post-1990 sample

Team 19 — *visibility-suppression*

Hypothesis Countries with lower Liberal democracy levels will exhibit a lower mean normalized citation ratio for SSH articles, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.

Outcome variable `log_cite_ratio_mean` — country-year mean of log normalized citation ratio

Predictor `v2x_libdem`

Controls `log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE

Robustness checks RC1 — Exclude recent articles; RC2 — Alternative regime measure; RC3 — Alternative normalization; RC4 — Aggregation floor sensitivity

Team 20 — *visibility-suppression*

Hypothesis The citation penalty associated with lower Liberal democracy levels will be larger for politically sensitive articles than for non-sensitive articles, as indicated by a positive coefficient on the v2x_libdem $\times$ sensitive_flag interaction term, after controlling for country fixed effects, year fixed effects, field fixed effects, log GDP per capita, and log population.

Outcome variable `log1p(cite_ratio)` — article-level log(1 + citation ratio); tests the v2x_libdem x topic-sensitivity interaction

Predictor `v2x_libdem`

Controls `log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE

Robustness checks RC1 — Alternative sensitivity flag; RC2 — Alternative regime measure

Team 21 — *visibility-suppression*

Hypothesis Countries with lower Liberal democracy levels will exhibit a lower share of SSH articles that receive at least one citation within a country-year, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.

Outcome variable `prop_ever_cited` — country-year share of articles with at least one citation

Predictor `v2x_libdem`

Controls `log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE

Robustness checks RC1 (sample restriction); RC2 (alternative regime measure)

Team 22 — *visibility-suppression*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a higher Gini coefficient of citations across SSH articles, after controlling for country fixed effects, year fixed effects, log GDP per capita, log population, and log article volume.
Outcome variable	`gini_tot_cites` — country-year Gini coefficient of tot_cites across articles
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`, `log(n_articles)`; plus country FE + year FE
Robustness checks	RC1 — Exclude recent articles; RC2 — Alternative regime measure; Alternative cell-size thresholds; Top-1% citation share; Normalized citations Gini; Lagged IV

Team 23 — *ideological-alignment*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a lower share of SSH abstracts with liberal/neutral political-economy framing (NEUTRAL + NONE) and a higher share with ideologically aligned framing (LEFT or RIGHT-CONSERVATIVE), after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_apolitical` — country-year share of NEUTRAL + NONE abstracts in the 5-label LLM ideology classifier
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE
Robustness checks	Alternative regime measure (binary); Alternative regime measure (ordinal); Post-2000 restriction; Fractional logit; Exclude NONE category

Team 24 — *ideological-alignment*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a higher share of SSH abstracts that actively legitimize or endorse the current political system, state authority, or leadership, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_legitimating` — country-year share of abstracts actively legitimating the domestic political system (LLM binary)
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE
Robustness checks	RC1 — Sensitive fields only; RC2 — Alternative regime measure; RC3 — Regime-type heterogeneity; RC4 — Abstract-presence check; RC5 — Fractional logit

Team 25 — *ideological-alignment*

Hypothesis	Countries with lower Liberal democracy levels will exhibit a higher share of SSH abstracts that actively frame liberal democracy, Western political institutions, or international human rights norms as fundamentally flawed or illegitimate, after controlling for country fixed effects, year fixed effects, log GDP per capita, and log population.
Outcome variable	`share_anti_liberal` — country-year share of abstracts with explicit anti-liberal-democracy framing (LLM binary)
Predictor	`v2x_libdem`
Controls	`log(e_gdppc)`, `log(e_wb_pop)`; plus country FE + year FE
Robustness checks	RC1 — Sensitive fields only; RC2 — Alternative regime measure; RC3 — Pre-2011 restriction; RC4 — Non-linear specification

D LLM classification prompts

[Full prompts for Teams 05, 10, 12, 23, 24, 25 — to be extracted from analysis plans.]

E Variable codebook

[Key variables from `agent_corpus.rds`. Cross-reference `data/vdem_codebook.md`.]